# Coding & Solution Implementation

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| **Template: Coding & Solution Quality Evaluation** | **Maximum Marks: 5 Marks** |
| **Date:** 15 March 2026 | **Team ID:** FV-AI-2026-X1 |
| **Project Name: FoodVerse AI** | **Track:** AI/ML & Generative AI |

## Coding & Solution:

*This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.*

## Solution Summary:

| Field | Implementation Details (FoodVerse AI) |
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| Project Title | FoodVerse AI — Multimodal Food Intelligence & Precision Nutrition Agent |
| Public GitHub Repository | https://github.com/AbhiragMishra/foodverse-ai (Latest Commit: aeb30da) |
| Live Production Deployment | https://foodverse-ai-six.vercel.app/ (Frontend on Vercel, Backend on Render) |
| Core Programming Languages | Python 3.10+, JavaScript (ES6+ / React 18.3), HTML5, CSS3 |
| Primary Frameworks & Libraries | FastAPI, Uvicorn, OpenCV (opencv-python-headless), NumPy, PIL, Pydantic v2, Vite 6 |
| Artificial Intelligence Models | Groq Cloud Qwen 3.8B Multimodal Vision Model, Whisper Large v3 Turbo |
| Automated Test Suite Status | 100% Passed (9/9 edge-case and robustness unit tests verified) |

## Core API Implementation Endpoints:

| HTTP Method | API Route | Controller Implementation | Functionality |
|---|---|---|---|
| POST | /api/food-recognition/analyze | routers/food_recognition.py | Performs visual plate decomposition, calorie counting, and macronutrient estimation. |
| POST | /api/freshness-detector/analyze | routers/freshness_detector.py | Runs OpenCV HSV discoloration analysis and Laplacian texture roughness scoring. |
| POST | /api/hygiene-monitor/audit | routers/hygiene_monitor.py | Extracts representative video frames to audit kitchen sanitation and PPE compliance. |
| POST | /api/kitchen-assistant/generate-recipe | routers/kitchen_assistant.py | Generates structured step-by-step recipes tailored to user ingredients. |
| POST | /api/kitchen-assistant/ask-assistant | routers/kitchen_assistant.py | Provides hands-free real-time culinary Q&A; assistance during active cooking. |
| POST | /api/meal-planner/generate-plan | routers/meal_planner.py | Calculates BMR/TDEE targets and creates personalized multi-day allergen-safe plans. |